

KRISTIAN SEEGER

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Research Profile

Photonics physicist with expertise in semiconductor photonic devices, nonlinear dynamics, and nanoscale light-matter interaction. My research spans analytical theory, numerical modeling, electromagnetic simulations, device design, and experimental characterization, with a focus on understanding and engineering nanophotonic systems.

Skills

- **Research expertise**
Semiconductor active devices • Nanophotonics • Non-Hermitian Photonics • Nonlinear dynamics • Nonlinear photonics
- **Theoretical methods**
Quasinormal modes • Coupled-mode theory • Dynamical system theory • k-p theory • Adiabatic approximations
- **Experimental methods**
Photonic chip design • Experimental optics and photonics • Automation of experimental setups • Data-analysis and visualisation
- **Scientific programming**
MATLAB • Python • COMSOL • Julia
- **Soft skills**
Presentation of scientific research • Scientific writing • Teaching • Supervision

Education and Research

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| Technical University of Denmark
<i>Postdoctoral researcher, Department of Electrical and Photonics Engineering</i> | Oct. 2025 – Aug. 2026
<i>Kgs. Lyngby, Denmark</i> |
| • Project: Modal properties of nanostructured lasers with Prof. Jesper Mørk | |
| Technical University of Denmark
<i>PhD, Department of Electrical and Photonics Engineering</i> | Mar. 2022 – Nov. 2025
<i>Kgs. Lyngby, Denmark</i> |
| • PhD thesis: Theory and experimental characterization of nanostructured lasers with Prof. Jesper Mørk | |
| Université Côte d'Azur
<i>Visiting PhD researcher, INPHYNI</i> | Nov. 2024 – Mar. 2025
<i>Nice, France</i> |
| • Project: Non-Hermitian photonics with coupled nanolasers with Prof. Fabrice Raineri, Dr. Guilhem Madiot | |
| Technical University of Denmark
<i>Master of Science in Engineering (Master of Science in Physics and Nanotechnology)</i> | Sep. 2019 – Feb. 2022
<i>Kgs. Lyngby, Denmark</i> |
| • Master's thesis: Theory and Simulation of Semiconductor Fano Lasers with Prof. Jesper Mørk | |
| Nanyang Technological University
<i>Exchange semester abroad</i> | Sep. 2018 – Dec. 2018
<i>Singapore</i> |
| Technical University of Denmark
<i>Bachelor of Science in Engineering (Bachelor Program in Physics and Nanotechnology)</i> | Sep. 2016 – Jun. 2019
<i>Kgs. Lyngby, Denmark</i> |
| • Bachelor's thesis: Subradiant Multiatom States for Quantum Memory with Prof. Martijn Wubs | |

Journal Publications

- **K. Seegert** & J. Mørk: *On the confinement factor and mode volume of microscopic lasers*, (in preparation)
- **K. Seegert**, Y. Yu, M. Heuck & J. Mørk: *Instantaneous modes in dispersive laser cavities*, Physical Review A **114**, 033507 (2026)
- **K. Seegert**, R. Gajardo, G. Huyet, F. Raineri & G. Madiot: *Dynamical control of non-Hermitian coupling between sub-threshold nanolasers enables Q-switched pulse generation*, ACS Photonics, **13**, 5083 (2026)
- J. Mørk, M. Xiong, **K. Seegert**, M. Marchal, et al.: *Nanostructured Semiconductor Lasers*, IEEE Journal of Selected Topics in Quantum Electronics, **31**, 1 (2025)
- **K. Seegert**, M. Heuck, Y. Yu & J. Mørk: *Self-pulsing dynamics in lasers with dispersive mirrors*, Physical Review A **109**, 063512 (2024)
- M. V. Gurrieri, E. V. Denning, **K. Seegert**, P. T. Kristensen, & J. Mørk: *Dynamics and condensation of polaritons in an optical nanocavity coupled to two-dimensional materials*, Physical Review B **109**, 155432 (2024)
- M. Rode, G. Meucci, **K. Seegert**, T. Kiørboe, & A. Andersen: *Effects of surface proximity and force orientation on the feeding flows of microorganisms on solid surfaces*, Physical Review Fluids **5**, 123104 (2020).

Conference presentations

- (Invited) **K. Seegert**, R. Gajardo (presenter), G. Huyet, F. Raineri & G. Madiot: *Dynamical control of non-Hermitian coupling between sub-threshold nanolasers enables Q-switched pulse generation*, European Optical Society Annual Meeting 2026
- **K. Seegert**, G. Madiot (presenter), G. Beaudoin, K. Pantzas, et al.: *Q-switching with phase-coupled nanolasers*, 33rd Annual Laser Physics Conference 2025
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Regimes of self-pulsing in lasers with dispersive mirrors*, The International Symposium on Physics and Applications of Laser Dynamics 2023
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Complex dynamics of lasers with passive dispersive reflectors*, Nonlinear Dynamics in Semiconductor Lasers 2023
- **K. Seegert**, Y. Yu, M. Heuck, & J. Mørk: *Dual-mode lasing and beating oscillations in a microscopic laser based on electromagnetically induced transparency*, 2023 Conference on Lasers and Electro-Optics Europe & European Quantum Electronics Conference
- **K. Seegert**, M. Heuck, Y. Yu, & J. Mørk: *Dispersive self-Q-switching in a microscopic laser*, IEEE Photonics Conference 2022

Teaching Experience

DTU Compute, DTU Physics, DTU Electro

Aug. 2017 – Aug. 2026

Kgs. Lyngby, Denmark

- **Teaching Assistant:** Advanced Engineering Mathematics 1 and 2 (01006, 01035), Physics 1 (10024), Model Physics (10050), Digital Signal Processing (62743), Nanophotonics (34053)
- **Supervision:** Master's projects and special courses in nanophotonics

Further Information

Other software: LaTeX • Word • Excel • PowerPoint • GDSPY • Emode Photonix • Inkscape • Maple • Mathematica

Languages: Danish (native) • English (fluent) • Spanish (basic) • German (basic)

Interests: Electronic music • Chess • Philosophy